Experiment 6 Day 3: Conversion of Borneol to Camphor

Alison Brown

Partner(s): None

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CHEM 2081-001

Introduction

This experiment was part three of the three-day process of converting camphene to camphor. For this third day, the goal was to convert borneol to camphor using the collected borneol product from the previous week, acetone, dimethyl chloride, glacial acetic acid, aqueous sodium bicarbonate solution, and sodium hypchloride. The preparation of camphor was done by allowing the borneol-bleach-acetic-acid-acetone reagent mixture to react in a hot water bath, then separating the camphor product through a series of washing and decanting.

Materials and Methods

After a week had passed, the borneol product that had been placed on the watch glass had unexpectedly evaporated. While the product had been white and powdery and had taken up much of the surface of the watch glass before it went into the drawer, there was now only a small dry stain left on the glass. However, another student had plenty of extra product to spare, so some of that borneol was used for this lab.

The sand bath was plugged into the 10 amp outlet and set to 60, then a beaker filled three-quarters of the way with water was placed in it to prepare a hot water bath. A racemic mixture of 0.3g of the borneol product and 0.3mL of glacial acetic acid was poured into the 10mL long-neck round-bottom flask. This mixture was then dissolved in 1mL of acetone, then a boiling chip was also added to the flask. A reflux apparatus was assembled by connecting a distillation column to the round-bottom flask, which was set in the water bath. The thermometer was set in the water as well to monitor that the water did not deviate from the temperature range between 40 and 50 degrees Celsius.

The borneol dissolved at this temperature, so there was no need to add an additional 0,5mL of acetone. Aqueous sodium hypochloride solution, also known as bleach, was added to

the mixture dropwise through the top of the condenser, incorporating a total of 6mL into the reagent solution. Each addition needed to be slow and careful, since the reaction was very volatile and could abruptly shoot up into the apparatus if too much bleach was introduced too quickly. The reaction was left to continue for 15 minutes following the last bleach addition, during which infrequent bubbles were observed and two layers formed. The apparatus was then removed from the bath to cool to room temperature.

From there, the solution was decanted into a centrifuge tube. The expected camphor product was extracted using two 2mL portions of methylene chloride since the two compounds have similar structures and would dissolve into one another. Dichloromethane has a density of 1.33g/mL, greater than water, so the organic layer was expected to be the bottom of the two layers. The top aqueous layer was pipetted out and collected in an Erlenmeyer flask just in case the incorrect layer had been removed. Any remaining glacial acetic acid present in the centrifuge tube was neutralized with the addition of about 2mL of sodium bicarbonate. After that, 2mL of 5% sodium bisulfate and 2mL of water were sequentially added to wash the organic mixture. The resulting aqueous layer was again pipetted out and into the Erlenmeyer flask.

Anhydrous sodium sulfate was then poured into the centrifuge tube to remove any microscopic traces of water remaining in the organic solution. Once the drying agent crystals were freely rolling at the bottom of the tube, the now-dry organic solution was poured onto a pre-weighed watch glass. The watch glass was then placed over the top of the water bath beaker to allow the dichloromethane to evaporate and leave behind the solid product crystals. After all the liquid on the watch glass had evaporated, the watch glass was weighed again.

Lab Performance

Exp 6 D5

Allison Brown — Name:
004 — Section:
— Room:
J — Hood:

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves. ○ ○ ○ ○ ~~○~~ (5pts)

Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks. ○ ○ ○ ○ ~~○~~ (5pts)

Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts). ○(1pt) ○(1pt) ○(1pt) ○ (1pt) ~~○ (5pts)~~

Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts). ○(1pt) ○ (5pts) ~~○ (5pts)~~

Working independently or equal partner participation during the entire lab. ○ ○ ○ ○ ○ ~~(5pts)~~

Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace. ○ ○ ○ ○ ○ ~~(5pts)~~

Maintain clean and organized personal space. ○ ○ ○ ○ ○ ~~(5pts)~~

GTA notes:

GTA signature: _____ *Final Performance Grade:* [redacted]

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Ex Day 3	Conversion of Borneol to Camphor	29
Name: Allison Brown	Lab Partner	Locker/Desk No. / Course & Section No.

Procedure:

1. In the long-neck 10mL round-bottom flask, add racemic 0.3g of your borneol product with 0.3mL of glacial acetic acid. Dissolve the racemic mixture in 1mL of acetone. Add a boiling chip
2. Just like in Day 2, assemble a reflux with the distillation column and place the apparatus in a water bath with the temperature between 40-50°C. Make sure the temperature of the water bath does not deviate from the specified range. If the borneol doesn't dissolve at this temperature, add 0.5mL of acetone to the mixture until all borneol goes into the solution
3. Once all reagents are dissolved, add 6mL of aqueous sodium hypochloride solution (bleach). The addition should be done through the top opening of the condenser and should be paced dropwise over 35 minutes. After each addition of the hypochloride, the mixture should be given a gentle swirl to ensure all chemicals are mixed thoroughly. The reaction must be allowed to continue for 15 minutes after the last bleach addition
4. Next, let the mixture cool to room temperature and extract the camphor with two portions (2mL each) of methylene chloride (or dichloromethane, or DCM). To make the extraction more efficient, use a centrifuge tube. ✓
5. Neutralize the combined methylene chloride layers with ~2mL of saturated sodium bicarbonate. Keep adding bicarbonate solution if necessary. The presence of acidic residuals will be determined by the production of CO_2 bubbles after every addition of the bicarbonate solution
6. Next, wash the organic solution sequentially with 2mL of 5% sodium bisulfite and ✓ with 2mL of water ✓

borneol product from last week no longer powdery, there was just a stain on the watch glass??? - was given extra borneol from Katie

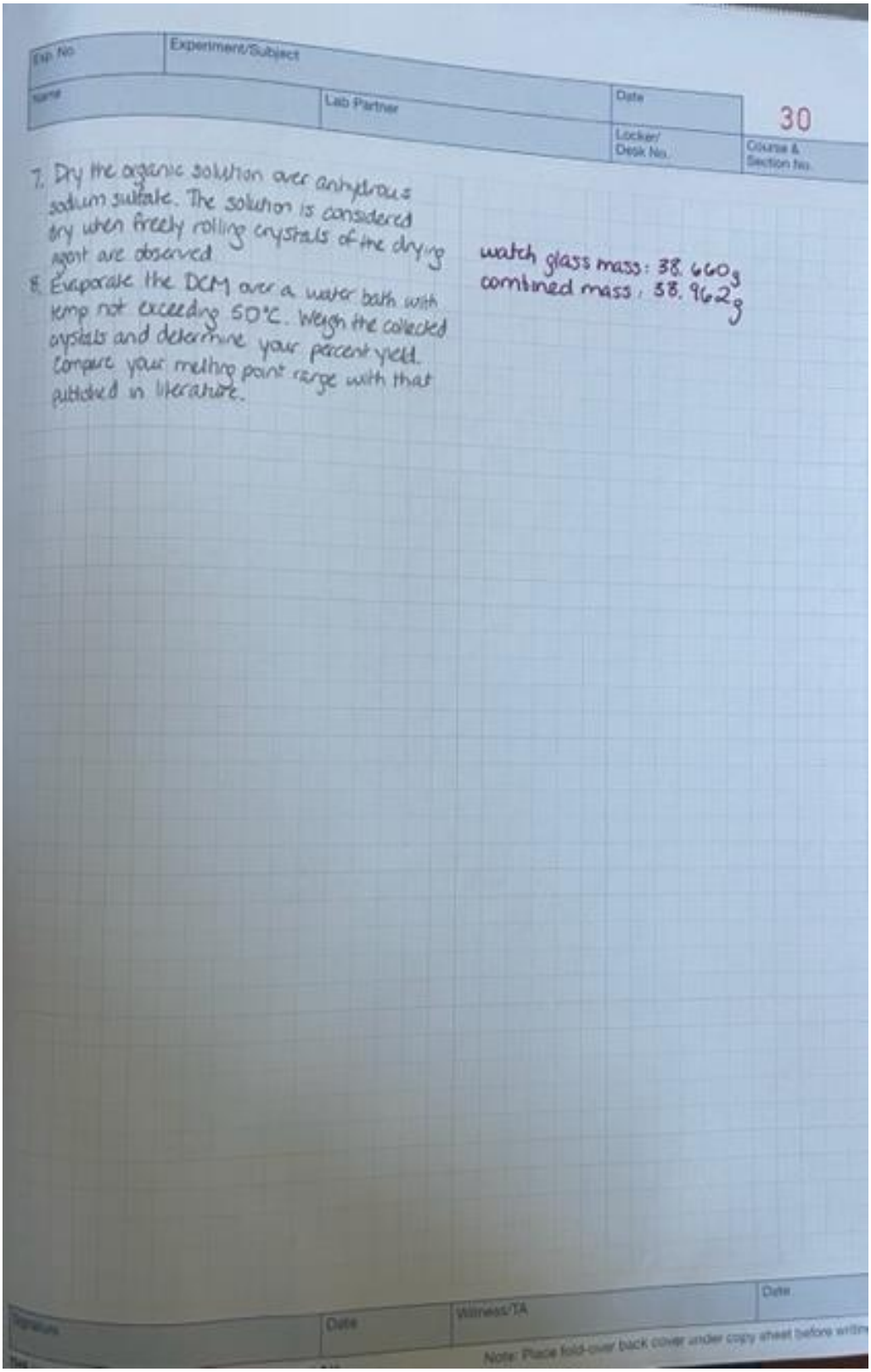
sand bath again is missing about 1/2 of the usual sand

temperature briefly got to 55°C, but hot water bath was taken out of sand bath to cool

after bleach addition, infrequent bubbles; two layers formed

Signature	Date	Witness/TA	Date

Note: Place fold-over back cover under copy sheet before writing



Exp No. | Experiment/Subject

Name | Lab Partner | Date | 30

Locker/ Desk No. | Course & Section No.

7. Dry the organic solution over anhydrous sodium sulfate. The solution is considered dry when freely rolling crystals of the drying agent are observed.
8. Evaporate the DCM over a water bath with temp not exceeding 50°C. Weigh the collected crystals and determine your percent yield. Compare your melting point range with that published in literature.

watch glass mass: 38.660 g
combined mass: 58.962 g

Signature | Date | Witness/TA | Date

Note: Place fold-over back cover under copy sheet before writing

Results

Watch Glass Mass	38.660g
Watch Glass and Collected Crystals Mass	38.962g
Mass of Collected Crystals	0.302g

Table 1. Determined mass of the collected camphor crystal product

Calculations

$$38.962g - 38.660g = 0.302g\ (mass\ of\ collected\ camphor\ crystals)$$

Theoretical Yield (1 mol borneol produces 1mol camphor)

$$0.3g * \frac{1mol}{154.25g} * \frac{1mol}{1mol} * \frac{152.23g}{mol} = 0.296g\ theoretical\ camphor\ mass$$

$$\frac{actual}{theoretical} * 100\% = \%\ yield \rightarrow \frac{0.302g\ actual}{0.296g\ theoretical} * 100\% = 1.02\%$$

Expected Mechanism

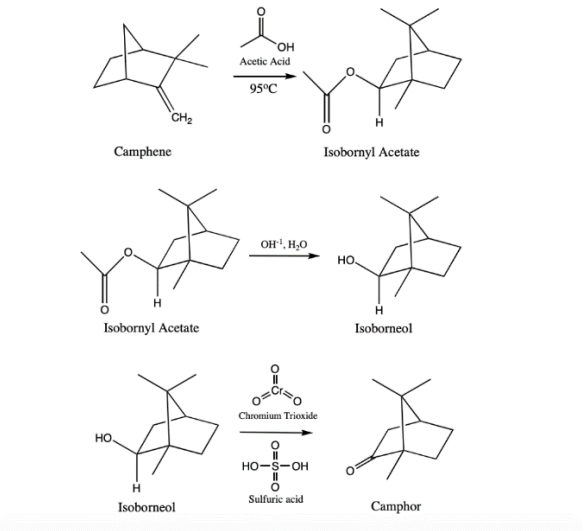


Figure 1. The overall mechanism of converting camphene to camphor

Discussion

In this experiment, about 0.302g of what was anticipated to be camphor was collected following the heating of the reaction, the separating of the resulting layers, and the evaporation of the dichloromethane. Based on this collected mass, and assuming the collected solid was the intended camphor product, the percent yield was 1.02%.

The borneol used in this lab was from another student's collected product, rather than the product collected from the previous week's lab. Surprisingly, despite the fact that the observed changes in the reaction and in the separation steps occurred as expected, the borneol product had evaporated after a week, leaving only a small stain on the watch glass rather than a white powder. A possible explanation for this product disappearance is that an incorrect layer was collected, so an aqueous layer remained and some of the organic product was unintentionally discarded. Any water present in the product following the vacuum filtration was supposed to evaporate over the week waiting period, and if some aqueous solution was mixed in with the collected product, it would have inflated the observed product, then evaporate out with the water. Had there been enough product to use for lab day three, potential errors for this lab would have been any other overlooked errors from the previous two lab days that would have compounded, or if there had been impurities in the previous product.

With the new product from another student, the third lab was able to proceed as expected. A thermometer was kept in the hot water bath to ensure that the reaction was kept within a range between 40 and 50 degrees Celsius – below that, and the reaction wouldn't complete, and above that, some of the reactants could evaporate. The bleach was carefully and slowly added dropwise, since the reaction was very volatile, and adding too much too quickly could cause the reaction rate to increase out of control and the solution to shoot up in the apparatus. Additionally,

the reaction was allowed to continue for a little over the recommended 15 minutes after the last addition of bleach. Another potential error for this lab could have been not allowing the mixture to cool to room temperature before disassembling the apparatus and pouring the contents into the reaction tube. If the temperature hadn't adequately dropped, some vapors could still be present in the apparatus and would be lost when the apparatus was taken apart.

As for the percent yield, the 0.02% excess mass could be accounted for following the dichloromethane evaporation. The watch glass was removed from the hot water bath after the crystals visibly appeared to be dry, but it's likely that some of the DCM hadn't fully evaporated and was still present for the second weighing of the watch glass.

References

Keller L, Joseph K, and Miguel S. 2022 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.